

SUNIX SDC Expansion Board

Serial Port Configuration on Linux

User Guideline

Driver Ver. 2.1.1.0

Doc Ver. 2.1

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1. Serial Port

Confirm that tools have been installed:

- **sdc-config**
- **sdc-uart-config**
- **snxterm**

If they are not installed, please check driver installation guideline and following the instructions.

1.1. Serial Port configuration

Serial port configuration tool which is provide user to configure the serial port setting, Include serial port interface switch, RS-422/485 bus terminate disable/enable, enter “sdc-config” to check the options.

`$sdc-config`

```
root@test-System-Product-Name:~# sdc-config
Usage: sdc-config [options] [args]

Options:
-d                Display all boards info
-D [ID]          Display board detail info
-S [device]      Serial port mode setting (-S help')
-O [ID] [q] [value] Store DIO output initial value ('-O help')
-U [ID] [file]   Update firmware ('-U help')
-V              Show driver version
-v             Show program version
root@test-System-Product-Name:~#
```

1.1.1. All boards info

Enter “sdc-config -d” to display all SDC boards info.

`$sdc-config -d`

```
root@test-System-Product-Name:~# sdc-config -d
ID0: model=CANE2104SI, firmware=1.14, pci_bus=3, irq=16
ID1: model=SDC4880B, firmware=1.9, pci_bus=4, irq=17
```

1.1.2. Board detail info

Enter “sdc-config -D {ID}” to display board detail info. {ID} = Board ID

```
$sdc-config -D {ID}
```

CAN

```
root@test-System-Product-Name:~# sdc-config -D 0
ID0: model=CANE2104SI, firmware=1.14, pci_bus=3, irq=16
CONFIG(0):
  ver=x00, pid=1, /dev=sdc_config0
  brand=1, model=0
CAN(1):
  ver=x00, pid=2, /dev=can0
  cap=x00000103, filters=32
  tx_msg_q_size=1024, tx_event_q_size=1024, rx_msg_q_size=4096
CAN(2):
  ver=x00, pid=3, /dev=can1
  cap=x00000103, filters=32
  tx_msg_q_size=1024, tx_event_q_size=1024, rx_msg_q_size=4096
CAN(3):
  ver=x00, pid=4, /dev=can2
  cap=x00000103, filters=32
  tx_msg_q_size=1024, tx_event_q_size=1024, rx_msg_q_size=4096
CAN(4):
  ver=x00, pid=5, /dev=can3
  cap=x00000103, filters=32
  tx_msg_q_size=1024, tx_event_q_size=1024, rx_msg_q_size=4096
```

Serial Port / DIO

```
root@test-System-Product-Name:~# sdc-config -D 1
ID1: model=SDC4880B, firmware=1.8, pci_bus=4, irq=17
CONFIG(0):
  ver=x00, pid=4, /dev=sdc_config1
  brand=1, model=0
UART(1):
  ver=x00, pid=5, /dev=ttyS4
  fifo=128, cap=rs232|rs422|rs485
UART(2):
  ver=x00, pid=6, /dev=ttyS5
  fifo=128, cap=rs232|rs422|rs485
UART(3):
  ver=x00, pid=7, /dev=ttyS6
  fifo=128, cap=rs232|rs422|rs485
UART(4):
  ver=x00, pid=8, /dev=ttyS7
  fifo=128, cap=rs232|rs422|rs485
DIO(5):
  ver=x00, pid=9, /dev=sdc_dio0
  cap=x03, nr_bank=2
  B0: nr_io=8, cap=input
  B1: nr_io=8, cap=output
```

```
root@test-System-Product-Name:~# sdc-config -D 1
ID1: model=SDC4880B, firmware=1.9, pci_bus=4, irq=17
CONFIG(0):
  ver=x00, pid=6, /dev=sdc_config1
  brand=1, model=0
UART(1):
  ver=x00, pid=7, /dev=ttyS4
  fifo=128, cap=rs232|rs422|rs485
UART(2):
  ver=x00, pid=8, /dev=ttyS5
  fifo=128, cap=rs232|rs422|rs485
UART(3):
  ver=x00, pid=9, /dev=ttyS6
  fifo=128, cap=rs232|rs422|rs485
UART(4):
  ver=x00, pid=10, /dev=ttyS7
  fifo=128, cap=rs232|rs422|rs485
DIO(5):
  ver=x01, pid=11, /dev=gpiochip1
  cap=x03, nr_bank=2
  B0: nr_io=8, cap=input
  B1: nr_io=8, cap=output
```

1.1.3. Serial port mode setting

Enter “sdc-config -S help” to inquire the serial port mode get/set syntax.

```
$sdc-config -S help
```

```

root@test-System-Product-Name: ~
File Edit View Search Terminal Help
root@test-System-Product-Name:~# sdc-config -S help
Usage: sdc-config -S [device] get|set [rs232|rs422|rs485] [term]

Example:
  Get mode (" -S ttyS4 get")
  Set mode (" -S ttyS4 set rs422 term")
  Set mode (" -S ttyS5 set rs485")
  term: bus terminate, for rs422,rs485
root@test-System-Product-Name:~#

```

- Change interface to RS-422.

```

root@test-System-Product-Name: ~
File Edit View Search Terminal Help
root@test-System-Product-Name:~# sdc-config -S ttyS4 get
ttyS4 config=rs232
root@test-System-Product-Name:~# sdc-config -S ttyS4 set rs422
Set ttyS4 config success
root@test-System-Product-Name:~# sdc-config -S ttyS4 get
ttyS4 config=rs422
root@test-System-Product-Name:~#

```

- Change interface to RS-485.

```

root@test-System-Product-Name: ~
File Edit View Search Terminal Help
root@test-System-Product-Name:~# sdc-config -S ttyS4 get
ttyS4 config=rs232
root@test-System-Product-Name:~# sdc-config -S ttyS4 set rs485
Set ttyS4 config success
root@test-System-Product-Name:~# sdc-config -S ttyS4 get
ttyS4 config=rs485
root@test-System-Product-Name:~#

```

1.1.4. Tool version

Enter “sdc-config -v” to show tool version.

Program Version

```
$sdc-config -v
```

```

root@test-System-Product-Name:~# sdc-config -v
Program version=2.1.1.0

```

Driver Version

```
$sdc-config -V
```

```

root@test-System-Product-Name:~# sdc-config -V
Driver version=2.1.1.0

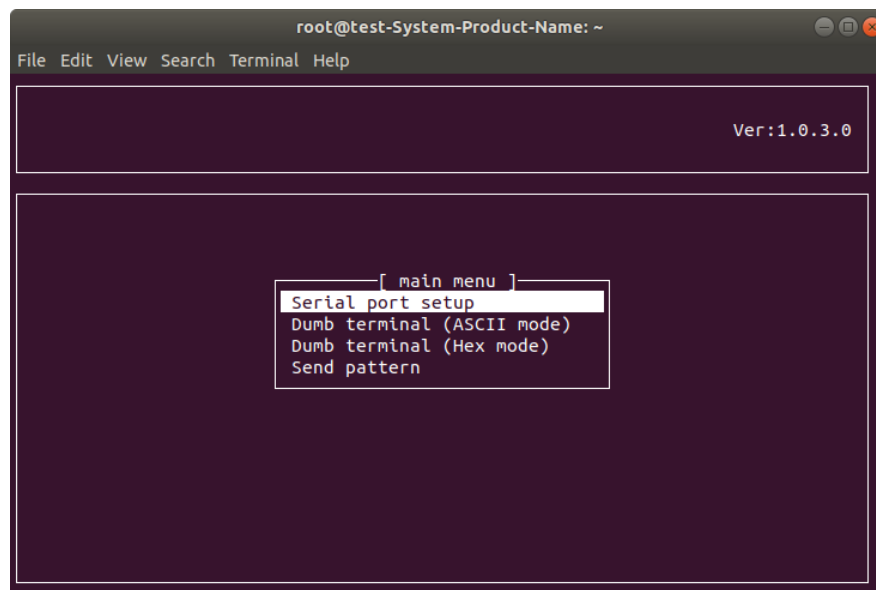
```

1.2. snxterm

snxterm is a terminal tool could test serial port communication, enter “snxterm” to execute the application.

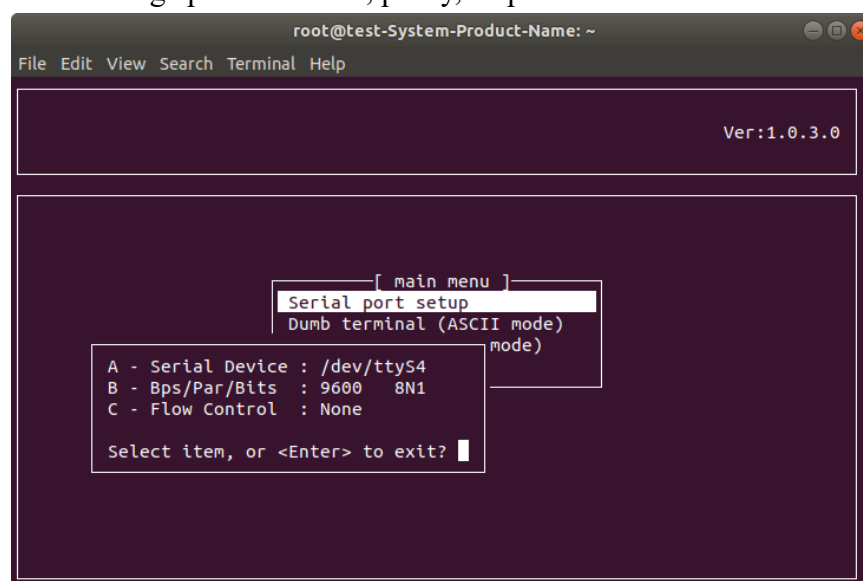
P.S. The following example is an RS232 Loopback test as an example.

\$snxterm



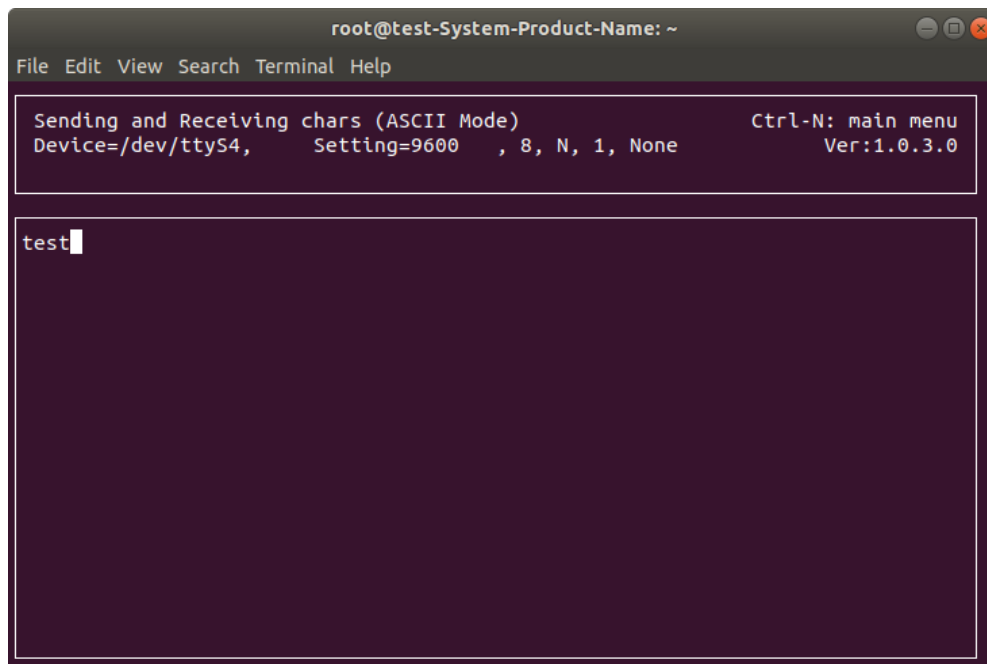
1.2.1. Serial port setup

Change port baud rate, parity, stop bits and flow control setting.



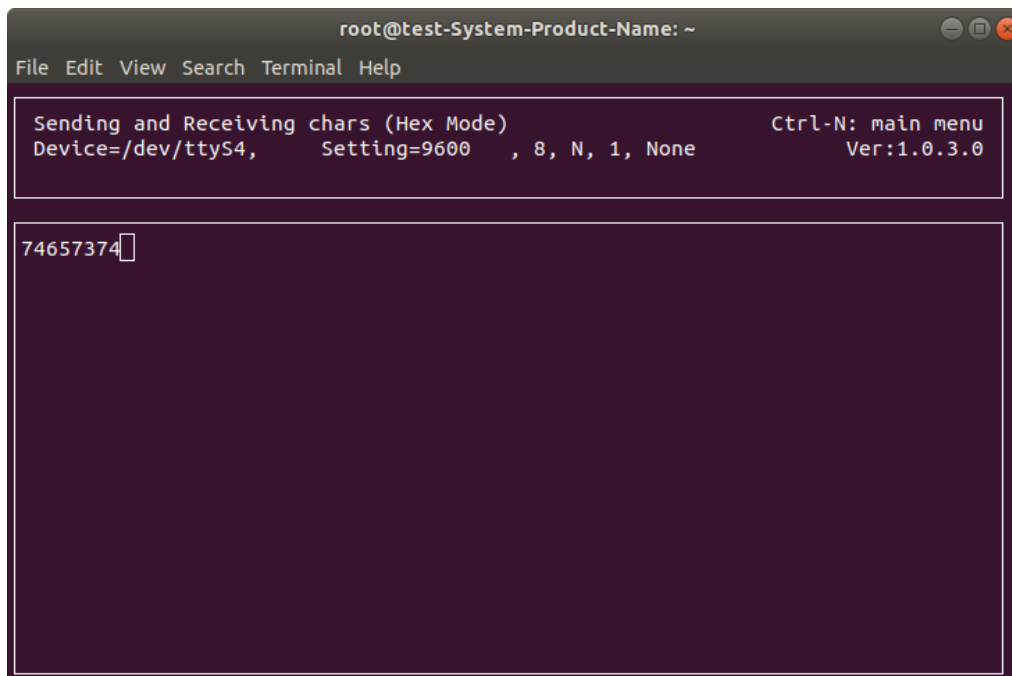
1.2.2. Dumb terminal (ASCII mode)

Enter any key, if the loopback connector plug correctly, the console should be show the received data (ASCII display).



1.2.3. Dumb terminal (Hex mode)

Enter any key, if the loopback connector plug correctly, the console should be show the received data (Hex display).



1.2.4. Send pattern

Send the preset string continuously.

```

root@test-System-Product-Name: ~
File Edit View Search Terminal Help

Sending and Receiving chars (Pattern Mode)                               Ctrl-N: main menu
Device=/dev/ttyS4,      Setting=9600      , 8, N, 1, None                Ver:1.0.3.0
Data length:40, Transmitting count:

[ default defined pattern ]
0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ

<Enter> to send      <Esc> to exit

```

```

root@test-System-Product-Name: ~
File Edit View Search Terminal Help

Sending and Receiving chars (Pattern Mode)                               Ctrl-N: main menu
Device=/dev/ttyS4,      Setting=9600      , 8, N, 1, None                Ver:1.0.3.0
Data length:40, Transmitting count:103

6789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123
456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123VW0123456789:;<=>?@ABCDEFGHIJKLMN0PQRS
TUVW0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQ
RSTUVW0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0
PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKL
NOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJK
LMNOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHI
JKLMNOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFG
HIJKLMNOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDE
FGHIJKLMNOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABC
DEFGHIJKLMNOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@A
BCDEFGHIJKLMNOPQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?
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:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ01234567
89:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ0123456789:;<=>?@ABCDEFGHIJKLMN0PQRSTUVWXYZ012345

```